EXPERIMENT – 08

LAPLACE TRANSFORM, INVERSE TRANSFORM, ROC AND STABILITY DATE

DATE: 23.06.26

OBJECTIVES :

ii) To determine the inverse Laplace transform of a given frequency-domain representation and reconstruct the corresponding time-domain signal.

iii) To determine the Region of Convergence (ROC) of Laplace transforms and analyze its significance in characterizing continuous-time signals and systems.

(ii) Determination of Inverse Laplace Transform for Signal Reconstruction.

PROCEDURE:

1. Define X(s) symbolically.
2. Apply ilaplace to obtain x(t).
3. Evaluate x(t) over $0 \leq t \leq 5$ s.
4. Plot the reconstructed time-domain signal. Observe plot
5. Compare the plotted response with the analytical expression.

MATLAB CODE:

MATLAB Coding clc; clear; close all;

Qualitative %% 8(ii) Determination of Inverse Laplace Transform for Signal Reconstruction

figure;

% Inverse Laplace Transform

% Requires Symbolic Math Toolbox

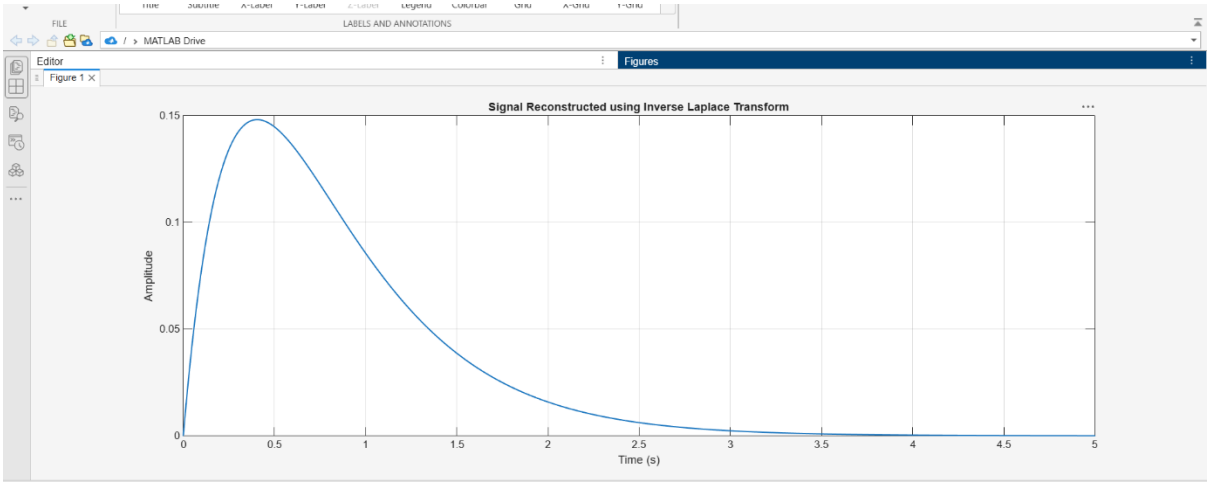
syms s t

X = 1/((s+2)*(s+3));

x = ilaplace(X,s,t);

```
disp('Inverse Laplace Transform x(t) =');
pretty(x);
tt = 0:0.001:5;
xx = double(subs(x,t,tt));
plot(tt,xx,'LineWidth',1.2);
grid on; xlabel('Time (s)');
ylabel('Amplitude');
title('Signal Reconstructed using Inverse Laplace Transform');
```

OUTPUT:



(iii) To determine the Region of Convergence (ROC) of Laplace transforms and analyze its significance in characterizing continuous-time signals and systems.

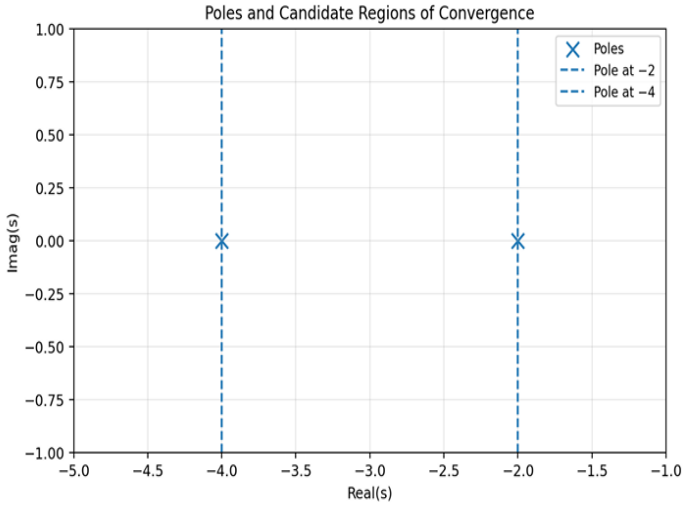
PROCEDURE:

1. Define the pole locations.
2. Plot the poles in the s-plane.
3. Mark the vertical lines through the poles. Record result
4. State the candidate ROC for each signal-sidedness case.
5. Interpret the relationship between poles and ROC.

MATLAB CODE:

```
clc; clear; close all;
%% 8(iii ) Analysis of Region of Convergence (ROC) and Its Significance in System Analysis
figure;
% ROC Analysis for a Rational Laplace Transform
% Requires Symbolic Math Toolbox
% X(s) = 1/((s+2)(s+4))
% Poles are at s = -2 and s = -4.
p = [-2 -4];
plot(real(p),imag(p),'rx','MarkerSize',10,'LineWidth',2);
hold on;
xline(-2,'--'); xline(-4,'--');
grid on; xlabel('Real(s)');
ylabel('Imag(s)');
title('Poles and Candidate Regions of Convergence');
legend('Poles','Pole at -2','Pole at -4');
fprintf('Poles are at s = -2 and s = -4.\n');
fprintf('For a right-sided signal, ROC: Re(s) > -2.\n');
fprintf('For a left-sided signal, ROC: Re(s) < -4.\n');
fprintf('For a two-sided signal, ROC: -4 < Re(s) < -2.\n');
```

OUTPUT:



CONCLUSION:

Thus, the inverse Laplace transform of the given frequency-domain representation was determined, and the corresponding time-domain signal was reconstructed. The Region of Convergence (ROC) was also analyzed to understand its significance in characterizing continuous-time signals and system stability.

LABORATORY PERFORMANCE ASSESSMENT RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks given	Marks obtained
(i) Determine the inverse Laplace transform of a given frequency-domain representation and reconstruct the corresponding time-domain signal.	**Inverse Laplace Transform Evaluation**	Correctly expands X(s) using partial fractions (distinct, repeated and complex poles) and applies the standard transform pairs and properties. Obtains the exact time-domain expression x(t), including the unit step u(t) and any shifts, with correct constants.	Applies partial fractions and transform pairs correctly, with minor errors in residues, constants or step-function terms.	Performs basic inverse transforms but has noticeable errors in partial fractions or transform pairs, and needs guidance.	Unable to determine the inverse Laplace transform.	**20**	**2**	
	Time-Domain Reconstruction & MATLAB Implementation	Uses suitable MATLAB commands (residue, ilaplace, syms, impulse or step) with a proper time vector. Reconstructs and plots x(t) with titles, axis labels and grid, and verifies the numerical result against the analytical expression.	Reconstructs and plots the signal correctly, with minor errors in code, time range, parameters or labelling.	Generates a basic reconstruction with noticeable errors in code or plots, and needs assistance.	Unable to reconstruct the time-domain signal or produce meaningful plots.	**20**	**2**	

(ii) Determine the Region of Convergence (ROC) of Laplace transforms and analyze its significance in characterizing continuous-time signals and systems.	**ROC Determination**	Correctly determines the ROC for right-sided, left-sided, two-sided and finite-duration signals from the pole locations. States the ROC as a proper inequality on Re(s) and shows it on the s-plane with poles and zeros marked.	Determines the ROC correctly for most signals, with minor errors in the boundary, signal type or s-plane plot.	Determines the ROC for basic cases only, with noticeable errors, and needs guidance.	Unable to determine the ROC of the given Laplace transforms.	**20**	**2**	
	ROC Significance for Signals and Systems	Clearly relates the ROC to the signal type, causality and BIBO stability (ROC contains the jω-axis), and shows that the same X(s) with different ROCs gives different time signals. Supports the reasoning with mathematical conditions and pole-zero evidence.	Relates the ROC to causality and stability correctly, with minor errors or limited justification.	Gives a basic explanation of the ROC's significance. Justification is incomplete and needs guidance.	Unable to explain the significance of the ROC.	**20**	**2**	
Interpret the Laplace-domain results and document the experiment.	**Engineering Analysis, Interpretation & Record**	Provides technically accurate interpretation of the results (pole locations, ROC and signal behaviour).	Maintains a proper record with generally accurate interpretation, with minor omissions.	Record and interpretation are basic and incomplete, with some technical inaccuracies.	Incomplete record, limited or incorrect interpretation of the results.	**20**	**2**	

		Maintains a complete, well-organized record with code, outputs, plots and conclusions.						
					Total	**100**	**10**	